AIRFOIL SELECTION & AERODYNAMIC ANALYSIS

Two RC aircraft missions: payload/stability (A) and maneuverability (B) | Submission-ready technical report

Abstract. This report compares three NACA four-digit sections for each mission and selects NACA 4412 as the starting candidate for payload/endurance (A) and NACA 0012 for aerobatic symmetry (B). NACA 2412 is retained as a compromise profile. Airfoil-level effects are separated from whole-aircraft design factors. Numerical values are identified as geometry-derived, analytical estimates, or sourced predictions; they are not presented as flight-test results.

1. Assumptions and candidate shortlist

Assumed foam RC aircraft; subsonic, low-speed flight; clean, fixed-wing configuration; chord c = 0.25 m and representative cruise speed V = 12 m/s for Reynolds-number estimation. With sea-level air ν ≈ 1.5×10■■ m²/s, Re = Vc/ν ≈ 2.0×10■. This is an illustrative analysis point; actual Re varies across the span and with speed. The airfoils are compared at matched Re in principle, but no XFOIL/XFLR5 executable was run to generate new polars for this report.

Mission A — maximum payload with stable flight: NACA 4412, NACA 2412, and NACA 23012. Mission B — maneuverability and rolls/loops: NACA 0012, NACA 0009, and NACA 0015. These are screening candidates, not a claim that any profile guarantees gentle stall or aerobatic capability.

2. Geometry and diagrams

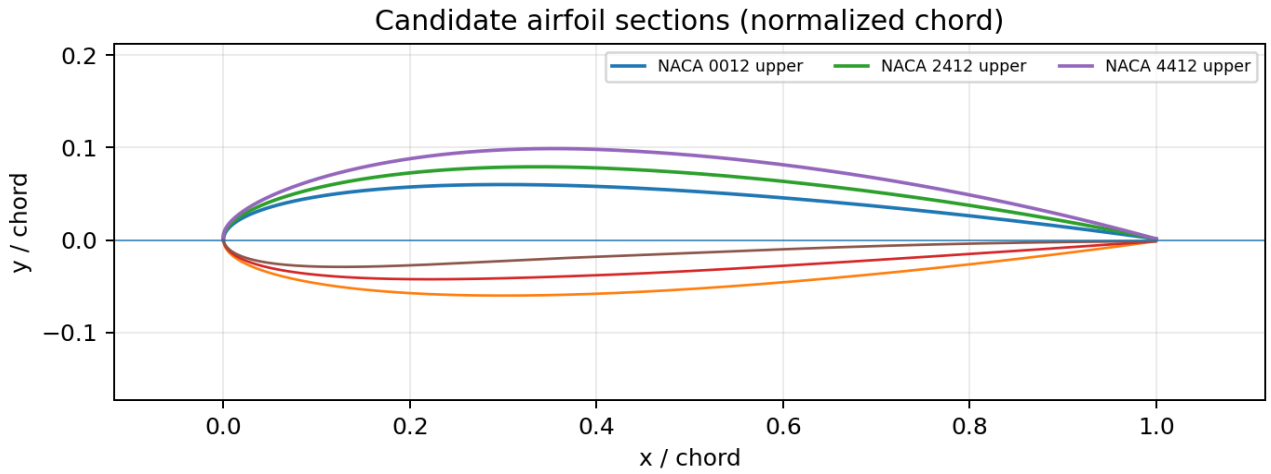


Figure 1. NACA four-digit section shapes normalized by chord. The plotted profiles are generated from the standard four-digit geometry equations; thickness is 12% for 0012/2412/4412. Geometry is schematic and not a manufacturing drawing.

For NACA four-digit profiles, digit 1 gives maximum camber (% chord), digit 2 gives the location of maximum camber in tenths of chord, and digits 3–4 give maximum thickness (% chord). Thus 0012 is symmetric (0% camber, 12% thickness); 2412 has 2% camber at 40% chord and 12% thickness; 4412 has 4% camber at 40% chord and 12% thickness. At c = 250 mm, 12% thickness = 30 mm; 2% camber = 5 mm; 4% camber = 10 mm. NACA 0009 and 0015 have 9% and 15% thickness, respectively.

3. Comparison of aerodynamic characteristics

Airfoil	Camber	Thickness	Lift / zero-lift	Drag and L/D	Stall / mission fit
NACA 0012	0%; symmetric	12% c	Cl ≈ 0 at α=0°; lift rises with α	No universal Cd advantage; evaluate at required Cl	Similar positive/negative behaviour; B candidate
NACA 2412	2% at 40% c	12% c	Positive lift at α=0°; αL=0 about −2°	Potentially efficient at moderate positive lift	Balanced compromise for A or general use
NACA 4412	4% at 40% c	12% c	Higher positive lift at low α; αL=0 more negative	Can improve useful L/D at positive cruise lift; verify polar	A candidate; more negative pitching moment, asymmetric inverted handling
NACA 23012	NACA 230-series camber distribution	12% c	Designed for useful lift; exact polar required	Candidate for lift-focused mission; compare at matched Re	A shortlist option; validate low-Re behaviour
NACA 0009 / 0015	0%; symmetric	9% / 15% c	Near-zero lift at α=0°	Thickness changes profile drag and structure; no assumed winner	B alternatives: thin section vs deeper/stiffer section

Lift coefficient Cl = L/(½ρV²S); drag coefficient Cd = D/(½ρV²S); section lift-to-drag ratio is Cl/Cd. These depend on angle of attack, Reynolds number, surface finish, transition, and separation. Camber shifts the zero-lift angle and changes pitching moment; thickness affects leading-edge radius, pressure recovery, internal volume, and structural depth. A higher Cl,max or a cambered section does not by itself prove lower drag or gentler stall.

4. Lift trend and coefficient interpretation

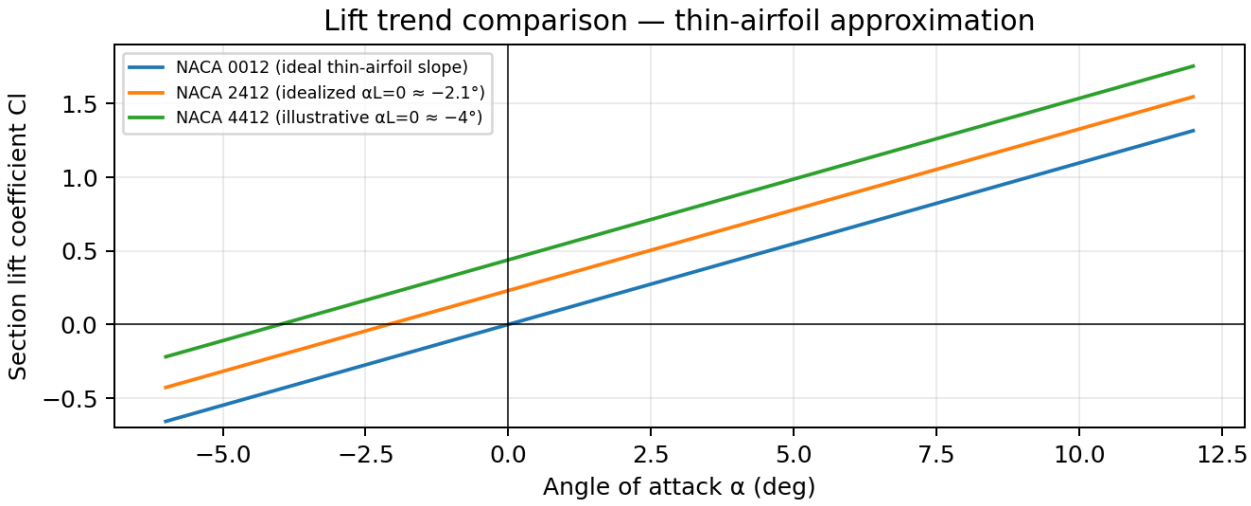


Figure 2. Thin-airfoil-theory trend only ($C_l \approx 2\pi(\alpha - \alpha_{L=0})$, radians). The NACA 2412 zero-lift angle is approximated from a representative published prediction; the NACA 4412 zero-lift angle shown is an illustrative camber-based approximation. This is **not** an XFOIL polar and does not predict stall, viscous drag, or low-Reynolds-number separation.

For payload flight, higher usable $C_{l,max}$ can reduce stall speed through $V_s = \sqrt{[2W/(\rho S C_{l,max})]}$, but wing area and mass are equally important. For maneuverability, symmetric sections allow more comparable upright and inverted lift response. A 4412 wing can still roll and loop; the trade-off is that inverted flight generally needs a less favourable angle of attack and trim.

5. Selection and mission trade-offs

Aircraft A — select NACA 4412 (12% thick, 4% camber at 40% chord). Its positive camber supports positive lift at low geometric angle of attack, making it a reasonable starting candidate for payload and endurance. Its pitching moment must be balanced by the tail/CG arrangement, and the selected profile should be checked at the actual cruise Reynolds number. NACA 2412 is the fallback if 4412's pitching moment or drag polar proves undesirable; NACA 23012 is another lift-focused candidate for simulation.

Aircraft B — select NACA 0012 (12% thick, symmetric). Symmetry gives near-zero lift at zero angle of attack and more similar upright/inverted lift characteristics. The 0009 is a thinner alternative, potentially reducing frontal thickness but providing less structural depth; 0015 offers more depth for structure but can carry additional profile drag. The final aerobatic choice depends on the whole aircraft's control authority, structural strength, and power-to-weight ratio.

Can one airfoil do both? Yes: NACA 2412 is a plausible compromise for a general-purpose RC aircraft, while 4412 prioritizes positive-lift cruise and 0012 prioritizes symmetric handling. A single section cannot simultaneously optimize every condition. Mixed root/tip sections are possible but should not be used without checking spanwise lift, pitching moment, and stall progression.

6. Whole-aircraft factors (not airfoil properties)

Wing loading is aircraft weight divided by planform area, W/S = mg/S. For an illustrative 1.2 kg aircraft with a 1.5 m span and 0.25 m rectangular chord, S = 0.375 m², mass loading = 3.2 kg/m² and weight loading ≈ 31.4 N/m². Lower wing loading generally supports slower flight, but does not guarantee higher efficiency. Stall speed also depends on the complete wing's Cl,max and configuration.

Thrust-to-weight ratio T/W affects acceleration and climb, not the section lift coefficient. A preliminary static target might be around 0.5–0.8 for endurance/payload, 0.8–1.0 for general-purpose flight, and 1.0–1.5 for aerobatics; these are planning ranges, not rules or proof of performance. Select motor, propeller, ESC and battery together, then check measured static thrust, current, temperature and all-up mass.

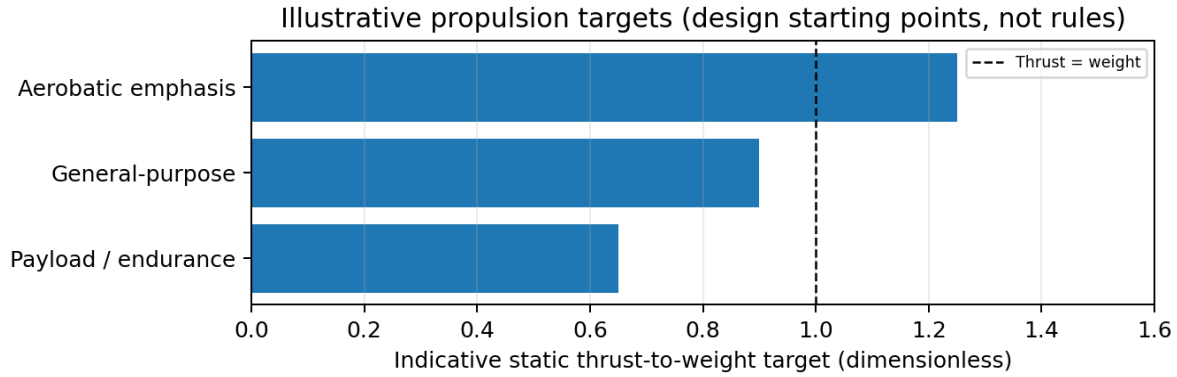


Figure 3. Illustrative thrust-to-weight targets for initial sizing; real requirements depend on propeller efficiency, airspeed, drag, climb demand and aircraft mass.

Other whole-aircraft factors include aspect ratio and planform (induced drag), aileron/elevator/rudder sizing and deflection (control authority), CG and tail volume (stability/trim), spar and joint strength (load factor), and surface finish/warping (transition and separation). An airfoil cannot compensate for an incorrectly placed CG or an underpowered motor.

7. Bonus analysis: matched-Re comparison and limitations

The representative condition is Re = 2.0×10■, corresponding approximately to c = 0.25 m and V = 12 m/s in air with ν = 1.5×10■■ m²/s. A proper bonus study should run all shortlisted sections in XFOIL/XFLR5 with the same Re, Mach number, Ncrit/transition assumptions, paneling and α sweep; export Cl–α, Cd–α, Cl–Cd and Cl/Cd plots; record convergence failures; and avoid treating solver failure near stall as a measured stall point. XFOIL predicts 2-D section behaviour and does not directly model finite-wing induced drag.

This report includes a transparent thin-airfoil-theory lift trend and geometry plot, but **does not claim a new XFOIL/XFLR5 simulation was run**. Published web summaries for NACA 0012 and 2412 provide useful context at Re = 200,000 (e.g., predicted Cl,max ≈ 1.12 and 1.35 respectively, under their stated model assumptions), but these are not a matched three-airfoil dataset and should not be mixed with a different model's 4412 numbers. Complete the matched polar run before claiming the +5 bonus is fully earned.

8. Conclusion

Use NACA 4412 as the initial payload/endurance candidate and NACA 0012 as the aerobatic candidate. Keep NACA 2412 as the compromise alternative. Confirm the selection using matched low-Re polars and a whole-aircraft assessment of wing loading, stability, control authority, structural margin and propulsion. Section-level coefficients must always be quoted with their Reynolds number, angle of attack and analysis method.

References

[1] University of Illinois Urbana–Champaign, *UIUC Airfoil Coordinates Database*, https://m-selig.web.engr.illinois.edu/ads/coord_database.html (coordinates and standard profile naming).
[2] XFOIL documentation, *Transition criterion and viscous analysis*, https://v0xnihili.github.io/xfoil-docs/analysis/ (Ncrit and boundary-layer assumptions).
[3] Foil.tools, *NACA 0012 polar summary*, https://foil.tools/foil/n0012 (modelled Re=200,000 summary; not experimental data).
[4] Foil.tools, *NACA 2412 airfoil summary*, https://foil.tools/foil/naca2412 (modelled Re=200,000 summary; not experimental data).
[5] NASA Technical Reports Server, *2-D lift and drag coefficients of the NACA 4412 airfoil at Re = 3,000,000*, https://ntrs.nasa.gov/api/citations/19750016605/downloads/19750016605.pdf (higher-Re data; not directly comparable to the selected RC condition).